

Primary Quadratic Modules for Heisenberg Fixed Fibres

Route-A structural certificate C156

Abstract

For a frozen Heisenberg lattice automorphism we determine the Smith type of every horizontal fixed-class group, reduce the universal central-rotation denominator to its group exponent, and split the resulting quadratic function into orthogonal primary components. Exact component ledgers through iterate fourteen turn the clean fixed-circle count into a product of primary zero counts. Primary is finite-group terminology here, not an arithmetic factorization.

Keywords: Heisenberg nilmanifold; finite quadratic module; Smith normal form; primary decomposition; fixed fibres; exact verification.

1 Smith form and the exponent bound

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $B = A^n$, and $M = B - I$. With F_n, L_n denoting Fibonacci and Lucas numbers, the standard doubling and Cassini identities give

$$M = \begin{cases} L_n \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}, & n \text{ odd}, \\ F_n \begin{pmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{pmatrix}, & n \text{ even}. \end{cases} \quad (1)$$

The cofactors have determinants -1 and -5 , respectively; the latter has content one for even n . Consequently

$$G_n = \mathbb{Z}^2 / M\mathbb{Z}^2 \cong \begin{cases} (\mathbb{Z}/L_n\mathbb{Z})^2, & n \text{ odd}, \\ \mathbb{Z}/F_n\mathbb{Z} \times \mathbb{Z}/(5F_n)\mathbb{Z}, & n \text{ even}. \end{cases} \quad (2)$$

Its exponent is $h_n = L_n$ for odd n and $h_n = 5F_n$ for even n .

For $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the canonical lattice correction is

$$q_B(x, y) = \frac{ac}{2}x(x-1) + bcxy + \frac{bd}{2}y(y-1). \quad (3)$$

The polarization of (3) is $(Bv)_1(Bw)_2 - v_1w_2$. The actual iterate correction has the same polarization, so $q_n = q_{A^n} + \ell_n$ with ℓ_n integer linear. This drift cannot be dropped when zero levels are counted. For $m = Mv$, define $\rho_n([m]) = q_n(v) - m_1v_2 \pmod{1}$.

Here is the parity step behind the improved denominator. Write $M = gU$, $U = \begin{pmatrix} r & s \\ s & t \end{pmatrix}$, where $(g, \det U, h) = (L_n, -1, L_n)$ in the odd branch and $(F_n, -5, 5F_n)$ in the even branch. By (2), $W = hv = (X, Y)$ is integral. Formula (3) gives

$$N := 2h\{q_{A^n}(v) - m_1v_2\} = X(m_2 - ac) - Y(m_1 + bd) + hm_1m_2 \in \mathbb{Z}. \quad (4)$$

Modulo two, division by five in the even branch is harmless. Substitution of $B = I + gU$ reduces (4) to

$$N \equiv s(gr + 1)(g + 1)X + s(gt + 1)(g + 1)Y + \{g(rt + s^2) + r + t\}XY. \quad (5)$$

Both Fibonacci and Lucas numbers have parity period 0, 1, 1. If $3 \mid n$, then $g = s = 0$ and $r = t = 1$; otherwise $g = 1$, $rt + s^2 = 1$, and $r + t = 1$. Thus (5) vanishes. Since $h\ell_n(v)$ is also integral, this proves

$$h_n\rho_n([m]) = 0 \quad \text{in } \mathbb{Q}/\mathbb{Z}. \quad (6)$$

This is an all-iterate divisibility result. Equality of the observed denominator with h_n is recorded only for $2 \leq n \leq 14$.

2 Orthogonal primary decomposition

If $u = Mw$, direct expansion gives the bilinear polarization

$$\beta_n([m], [u]) = v_1u_2 - u_1v_2 + m_1u_2 \pmod{1}. \quad (7)$$

Thus primary components of G_n at distinct primes are orthogonal, and the restriction to a component of exponent p^e takes values in $p^{-e}\mathbb{Z}/\mathbb{Z}$. Uniqueness of primary torsion in $\mathbb{Q}/\mathbb{Z}$ gives

$$C_n = \prod_{p|h_n} C_{n,p}, \quad C_{n,p} = \frac{1}{p^e} \sum_{a \bmod p^e} \sum_{x \in G_{n,p}} e^{2\pi i a \rho_n(x)}. \quad (8)$$

The formula is a finite root-of-unity zero projector, not an operator trace formula or an arithmetic local factor.

3 Exact certificate and boundary

CRT idempotents applied to the two standard generators, followed by column Hermite reduction, enumerate each primary component exactly. The counts are

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
C_n	1	1	4	1	21	4	57	1	148	105	397	144	1041	57

The producer, direct-cocycle checker, and SymPy path independently verify the ledger. The strict tuple is (A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT). Fixed sets remain clean circles and the ordinary isolated stability factor remains singular. We claim no isolated primitive-orbit determinant, target divisor, target functional equation or counting law, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.